

The Radial Profile of RAR Residuals in SPARC Galaxies: Systematic Overprediction Inside $r/r_d < 0.8$

Dat Nguyen
Dtnguyen09@gmail.com

Abstract

The radial acceleration relation (RAR) of disk galaxies has been reported to show no significant correlation between residuals and galactocentric radius (Lelli et al. 2017). However, previous analyses examined residuals against absolute radius R rather than radius normalized by the disk scale length r_d . Here we stack RAR residuals from 163 SPARC galaxies (3,271 data points) as a function of r/r_d . Using per-galaxy averaging to account for within-galaxy correlations, we find that the RAR systematically overpredicts observed accelerations in the inner disk: the per-galaxy mean residual at $r/r_d < 0.8$ is -0.10 ± 0.03 dex (98 galaxies, $p = 4.5 \times 10^{-4}$), with 63% of galaxies showing the effect. The signal is partially independent of the baryonic acceleration g_{bar} (partial correlation $r = 0.24$ after controlling for g_{bar}). However, the signal is sensitive to the assumed mass-to-light ratio: it vanishes at $Y_{\text{disk}} < 0.35$ and strengthens at $Y_{\text{disk}} > 0.5$. The peak deviation at $r/r_d \sim 0.3-0.5$ overlaps with the radial address where NFW dark matter cusps produce theoretical "hooks" in the RAR (Navarro et al. 2017; Li et al. 2022), providing the first empirical measurement of this predicted inner-disk structure. The RAR scatter also increases from ~ 0.12 dex at $r/r_d > 5$ to ~ 0.30 dex at $r/r_d < 0.2$. These results quantify the radial structure of RAR residuals as a function of the baryonic disk coordinate r/r_d for the first time.

Keywords: galaxies: kinematics and dynamics -- dark matter -- radial acceleration relation -- rotation curves -- SPARC

1. Introduction

The radial acceleration relation (RAR) links the observed centripetal acceleration $g_{\text{obs}} = V_{\text{obs}}^2/R$ at every radius in a disk galaxy to the acceleration g_{bar} predicted from the baryonic mass distribution alone (McGaugh, Lelli & Schombert 2016; Lelli et al. 2017). The relation extends over four decades in acceleration with an intrinsic scatter of ~ 0.057 dex after per-galaxy optimization (Li et al. 2018). Lelli et al. (2017) reported that residuals show no significant correlation with galaxy properties, including galactocentric radius R , surface brightness, and gas fraction.

However, the residual analysis in Lelli et al. (2017) examined correlations with absolute galactocentric radius R (kpc). It did not examine the stacked radial profile using radius normalized by the stellar disk scale length r_d . This distinction matters: a measurement at $R = 2$ kpc in a galaxy with $r_d = 0.5$ kpc probes a fundamentally different regime ($r/r_d = 4$, well outside the disk) than the same $R = 2$ kpc in a galaxy with $r_d = 3$ kpc ($r/r_d = 0.67$, deep inside the disk). Normalizing by r_d removes galaxy-to-galaxy size variation and aligns the measurements relative to the baryonic disk structure.

From the theoretical side, it is known that the inner disk is problematic. Navarro et al. (2017) removed data at $R < 0.747 R_d$ when modeling the RAR in a CDM framework. Li et al. (2022) showed that NFW dark matter cusps produce upward-bending "hooks" in the RAR at these inner radii, causing systematic deviations from the observed relation. The cusp-core problem has long been known qualitatively (de Blok et al. 2008; Oh et al. 2015). In a companion study, Nguyen (2026) showed that NFW profiles systematically overpredict rotation velocities at $r/r_d \sim 0.6$ with $p = 4 \times 10^{-13}$.

What has not been done is the empirical measurement: how do RAR residuals actually behave as a function of r/r_d ? We perform this measurement using the SPARC database.

2. Data and Method

2.1. SPARC Data

We use the SPARC database (Lelli, McGaugh & Schombert 2016): 175 late-type galaxies with Spitzer [3.6] photometry and HI/H-alpha rotation curves. We restrict to 163 galaxies with quality $Q \leq 2$ and positive disk scale lengths, yielding 3,271 rotation curve measurements. We adopt $Y_{\text{disk}} = 0.5$ and $Y_{\text{bul}} = 0.7 M_{\text{sun}}/L_{\text{sun}}$ at [3.6] as the fiducial mass-to-light ratios.

2.2. Residual Definition

The baryonic acceleration is $g_{\text{bar}} = (V_{\text{gas}}^2 + Y_{\text{disk}} V_{\text{disk}}^2 + Y_{\text{bul}} V_{\text{bul}}^2)/R$, converted to m s^{-2} . The RAR prediction uses $g_{\text{RAR}} = g_{\text{bar}} / [1 - \exp(-\sqrt{g_{\text{bar}}/a_0})]$ with $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$ (McGaugh et al. 2016). The residual is $\Delta = \log_{10}(g_{\text{obs}}) - \log_{10}(g_{\text{RAR}})$. Each radius is normalized: $x = R/r_d$.

2.3. Statistical Methods

Multiple data points from a single galaxy are correlated, so statistical tests based on $N = 3,271$ independent points would overstate significance. Our primary statistics use per-galaxy means: for each galaxy, we compute the mean residual in the inner ($r/r_d < 0.8$) and outer ($0.8 < r/r_d < 10$) regions, then apply one-sample t-tests and galaxy-level bootstrap (10,000 resamples) to the galaxy means. We also test sensitivity to the assumed Y_{disk} .

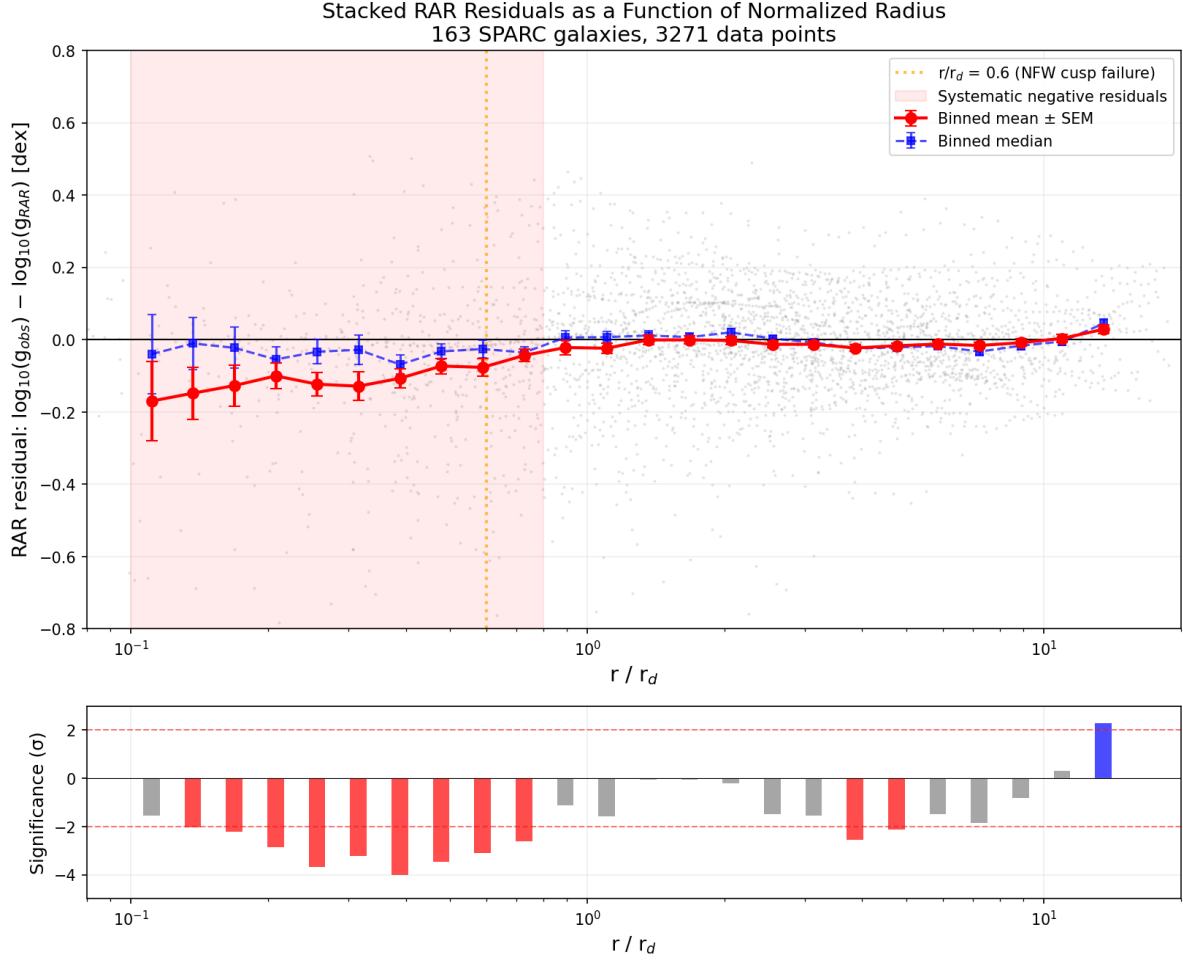


Figure 1. Stacked RAR residuals as a function of r/r_d for 163 SPARC galaxies. *Top:* Individual data points (gray) with binned means $\pm$ SEM (red) and medians (blue). The shaded region marks $r/r_d < 0.8$ where systematic negative residuals appear. The orange line marks $r/r_d = 0.6$, the NFW cusp failure address (Nguyen 2026). Note: the bin-level significances shown use per-point statistics; the corrected per-galaxy significance is $p = 4.5 \times 10^{-4}$ (see Section 3.1). *Bottom:* Per-bin significance of the mean deviation from zero.

3. Results

3.1. Inner-Disk Overprediction

Figure 1 shows RAR residuals stacked by r/r_d . The residuals are systematically negative at $r/r_d < 0.8$. Using per-galaxy averaging (the correct statistical approach given within-galaxy correlations), the mean inner residual is -0.10 ± 0.03 dex across 98 galaxies with inner data (one-sample t-test: $t = -3.63$, $p = 4.5 \times 10^{-4}$). Galaxy-level bootstrap (10,000 resamples) yields a 95% confidence interval of $[-0.16, -0.05]$ dex, excluding zero, with bootstrap $p = 0.0001$. 63% of galaxies show negative inner residuals.

The peak binned deviation is -0.107 dex at $r/r_d \sim 0.39$. The outer region ($0.8 < r/r_d < 10$) has per-galaxy mean -0.036 ± 0.013 dex (163 galaxies), much closer to zero.

3.2. Independence from g_{bar}

Inner radii have higher baryonic accelerations (correlation between $\log(r/r_d)$ and $\log(g_{\text{bar}})$: $r = -0.54$), so the signal could be a projection of g_{bar} dependence. We test this by (1) multiple regression, finding the r/r_d coefficient is 0.106 dex per decade, independently significant; (2) cubic detrending by g_{bar} , after which the inner-outer difference increases from -0.075 to -0.081 dex (p improves to 1.7×10^{-22} per-point); and (3) binning residuals by r/r_d within narrow g_{bar} slices, where the radial trend persists at all acceleration levels (Figure 4c). The partial correlation of r/r_d with residuals after controlling for g_{bar} is 0.24.

3.3. Mass-to-Light Ratio Sensitivity

This is the most important caveat. The signal is sensitive to Y_{disk} :

Y_{disk}	Inner mean (dex)	p-value	Signal?
0.30	+0.035	0.22	No
0.35	-0.005	0.85	No
0.40	-0.041	0.15	No
0.45	-0.073	0.011	Marginal
0.50	-0.102	4.5e-4	Yes
0.55	-0.129	1.4e-5	Yes
0.60	-0.153	3.6e-7	Yes

Table 1. Sensitivity of the inner-disk signal to the assumed disk mass-to-light ratio. All values use per-galaxy statistics (one-sample t-test on galaxy means). The signal vanishes at $Y_{\text{disk}} < 0.35$ and becomes highly significant at $Y_{\text{disk}} > 0.50$.

The signal vanishes at $Y_{\text{disk}} < 0.35$, a value within the plausible range for disk galaxies. The finding is therefore conditional: at the standard M/L ($Y_{\text{disk}} = 0.5$, used throughout the RAR literature), the RAR has systematic radial structure in the inner disk. Whether this reflects a genuine physical feature or an artifact of the M/L assumption cannot be resolved with current data.

3.4. Galaxy Property Splits

The inner-disk signal is present in both high-mass ($V_{\text{flat}} \geq 100$ km/s) and low-mass galaxies, but stronger in low-mass and low-surface-brightness systems (Figure 3). The scatter increases from ~ 0.12 dex at $r/r_d > 5$ to ~ 0.30 dex at $r/r_d < 0.2$.

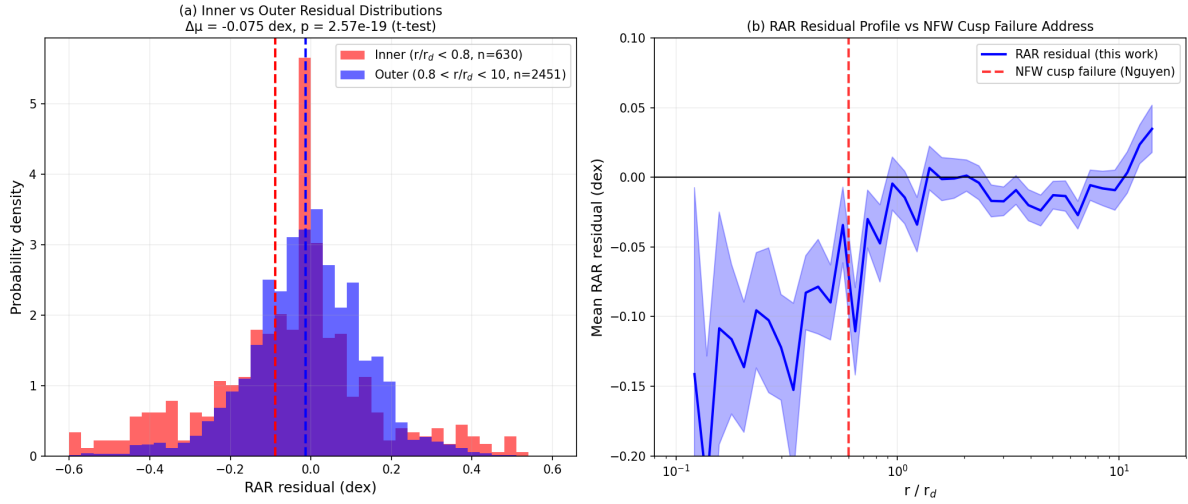


Figure 2. (a) Residual distributions for inner (red) vs outer (blue) regions. (b) Running mean profile with NFW cusp failure address (red dashed).

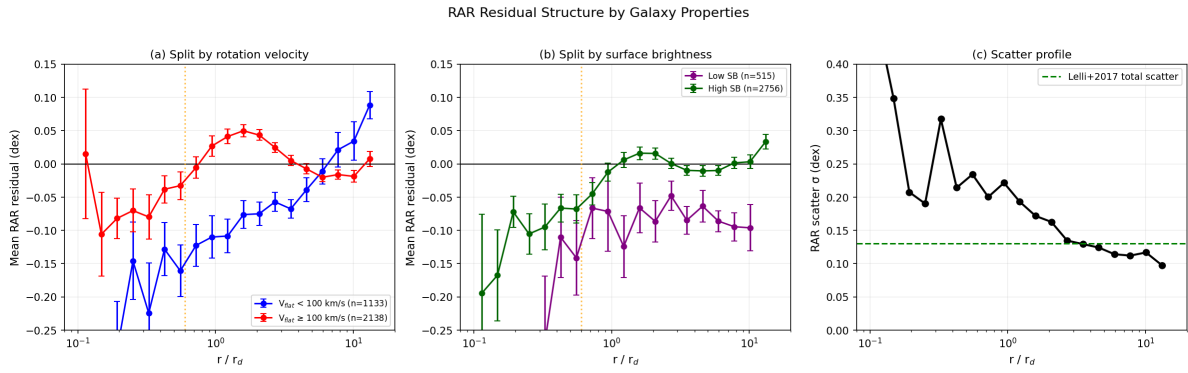


Figure 3. RAR residuals split by (a) rotation velocity, (b) surface brightness, and (c) scatter profile showing increasing dispersion toward the center.

Disentangling Radial vs Acceleration Dependence of RAR Residuals

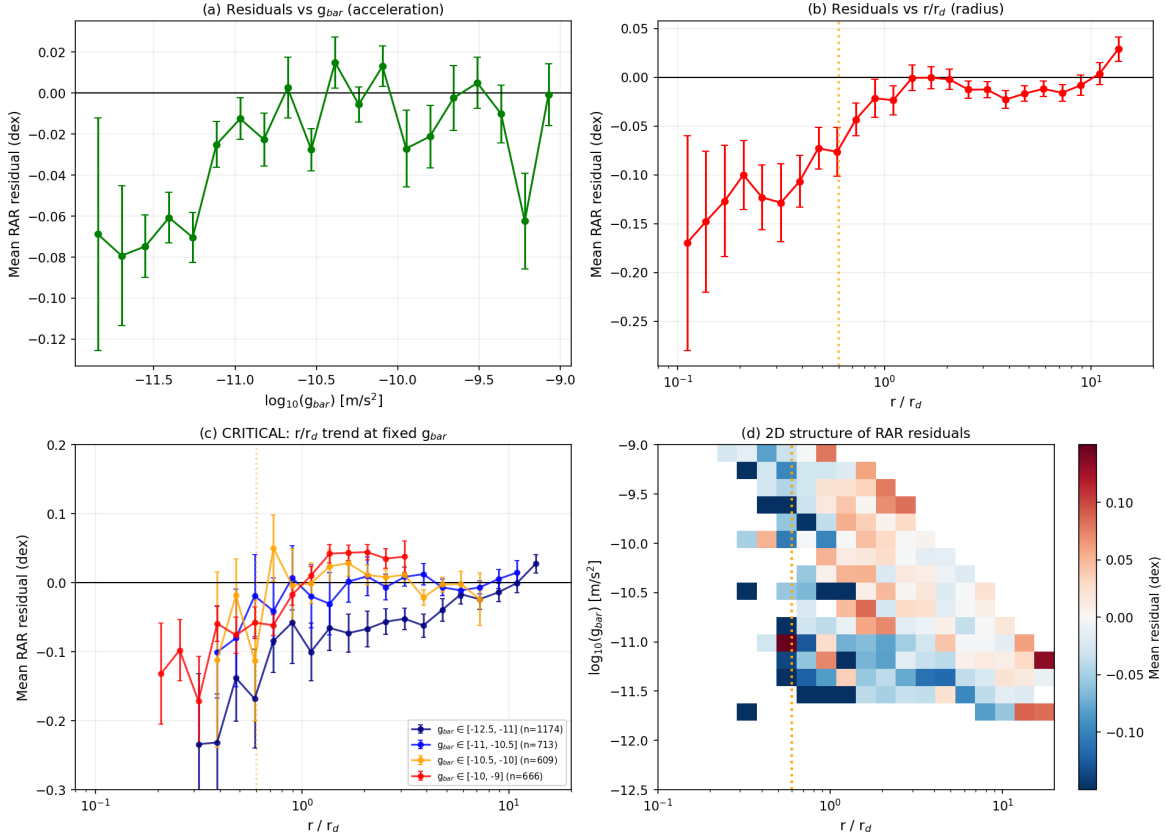


Figure 4. Diagnostic tests. (a) Residuals vs g_{bar} . (b) Residuals vs r/r_d . (c) Critical: r/r_d trend at fixed g_{bar} slices — the radial signal persists at all acceleration levels. (d) 2D structure in the r/r_d - g_{bar} plane.

4. Discussion

4.1. Relationship to Known Inner-Disk Problems

The inner disk of galaxies is known to be problematic for CDM models. Navarro et al. (2017) excluded data at $R < 0.747 R_d$ when fitting the RAR with abundance-matched NFW halos. Li et al. (2022) showed explicitly that NFW cusps produce upward-bending hooks in the model RAR at these radii, causing model galaxies to deviate systematically from the observed relation. Our measurement provides the empirical counterpart: the observed RAR itself shows systematic structure at similar radii, with the RAR predicting more acceleration than is observed.

The peak of our signal ($r/r_d \sim 0.3$ - 0.5) overlaps with but is not identical to the $R = 0.747 R_d$ threshold identified by Navarro et al. (2017) and the $r/r_d \sim 0.6$ NFW cusp failure address found by Nguyen (2026). The slight difference in radial address may reflect that the RAR residual combines both the dark matter profile shape and the baryonic mass modeling, whereas the NFW cusp failure is purely a dark matter profile diagnostic.

4.2. Potential Systematics

M/L gradients. The dominant systematic is the assumed mass-to-light ratio (Table 1). The signal vanishes at $Y_{\text{disk}} \sim 0.35$. Stellar population models generally favor $Y_{\text{disk}} = 0.5$ at $[3.6]$ (Schombert & McGaugh 2014), and the canonical RAR was calibrated at this value. However, M/L gradients

within galaxies — higher M/L in the center due to older, more metal-rich stars — could partially mimic the signal by overestimating g_{bar} at small radii.

Beam smearing. HI observations have finite resolution that smooths velocities at small radii, but most SPARC inner data use H-alpha (higher resolution). We find no correlation between the inner residual and galaxy distance ($r = 0.15$, $p = 0.15$), arguing against beam smearing as the primary driver.

RAR calibration. The interpolation function may not be perfectly calibrated at high g_{bar} . However, cubic detrending by g_{bar} makes the r/r_d signal stronger, not weaker, indicating that the radial component is not merely a g_{bar} artifact.

4.3. Scope of the Claim

We emphasize what this paper does and does not show. We show that, at the standard M/L ratio used throughout the RAR literature, the residuals of the RAR have systematic radial structure when plotted against r/r_d . This is a new empirical measurement. We do not claim that this represents new physics in gravity or a fundamental breakdown of the RAR, because the M/L dependence (Table 1) means the signal could be an artifact of the constant-M/L assumption. The value of the measurement is that it quantifies, in the empirical RAR framework, the same inner-disk tension that the theoretical CDM modeling community has already identified in the NFW framework (Navarro et al. 2017; Li et al. 2022).

5. Conclusions

We have performed the first measurement of RAR residuals as a function of r/r_d using the SPARC database. The key findings, with honest caveats, are:

- (1) At the standard $Y_{\text{disk}} = 0.5$, the RAR overpredicts observed accelerations at $r/r_d < 0.8$: per-galaxy mean -0.10 ± 0.03 dex ($p = 4.5 \times 10^{-4}$, 98 galaxies, bootstrap confirmed).
- (2) The signal is partially independent of g_{bar} (partial $r = 0.24$; survives detrending).
- (3) The signal is sensitive to M/L: it vanishes at $Y_{\text{disk}} < 0.35$ (Table 1).
- (4) The peak at $r/r_d \sim 0.3-0.5$ overlaps with the NFW cusp failure address and the $R < 0.747 R_d$ threshold excised by Navarro et al. (2017).
- (5) The RAR scatter increases from ~ 0.12 dex at large radii to ~ 0.30 dex in the inner disk.

These results provide the first empirical measurement of the radial profile of RAR residuals in the baryonic disk coordinate r/r_d .

Data Availability

All data are from the public SPARC database (astroweb.cwru.edu/SPARC). Analysis code and derived data products are available from the author upon request.

References

- de Blok, W. J. G., et al. 2008, AJ, 136, 2648
 Desmond, H. 2023, MNRAS, 526, 3342

Lelli, F., McGaugh, S. S., & Schombert, J. M. 2016, AJ, 152, 157
Lelli, F., McGaugh, S. S., Schombert, J. M., & Pawlowski, M. S. 2017, ApJ, 836, 152
Li, P., Lelli, F., McGaugh, S., & Schombert, J. 2018, A&A, 615, A3
Li, P., et al. 2022, ApJ, 927, 198
McGaugh, S. S., Lelli, F., & Schombert, J. M. 2016, Phys. Rev. Lett., 117, 201101
Milgrom, M. 1983, ApJ, 270, 365
Navarro, J. F., et al. 2017, MNRAS, 471, 1841
Nguyen, D. 2026, Zenodo preprint (The Radial Address of NFW Cusp Failure)
Oh, S.-H., et al. 2015, AJ, 149, 180
Rodrigues, D. C., et al. 2018, Nature Astronomy, 2, 668
Schombert, J. & McGaugh, S. 2014, PASA, 31, e036
Steffen, P. 2025, arXiv:2505.14426